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Outcomes of Second-Line Intravitreal Anti-VEGF Switch Therapy in Radiation Retinopathy Secondary to Uveal Melanoma: Moving from Bevacizumab to Aflibercept

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Keywords
Radiation retinopathy · Aflibercept · Bevacizumab · Anti-vascular endothelial growth factor · Uveal melanoma

Abstract
Introduction: Radiation retinopathy is a dose-dependent complication of the retina following exposure to ionizing radiation. The objective of this prospective case series is to determine the clinical efficacy of intravitreal aflibercept for radiation retinopathy secondary to radiotherapy for uveal melanoma in those that failed intravitreal bevacizumab treatment. ***Methods:*** A case series of 30 patients with a mean age of 57 ± 15 years with radiation retinopathy were enrolled. Visual acuity (VA) and central foveal thickness (CFT) responses to therapy were assessed with regression analyses at 1 month, 3 months, and 6 months following the switch to aflibercept. ***Results:*** Regression analyses showed a statistically significant reduction in CFT and improvements in VA following the switch to treatment by aflibercept at 1 month, 3 months, and 6 months. The mean CFT improved from 476 μm ± 170 to 386 μm ± 139 and the mean VA improved minimally from 20/115 ± 20/63 to 20/112 ± 20/54 over 6 months. After 6 months of aflibercept, 46% of patients displayed a CFT improvement of 100 μm or greater and 23% of patients showed improvement in VA of 1 line or better. ***Conclusion:*** This pilot study suggests that patients with radiation retinopathy who have failed monthly intravitreal bevacizumab may respond to aflibercept.

Introduction

Uveal melanoma is the most common primary malignancy of the eye and includes iris, choroidal, and ciliary body melanoma [1]. Enucleation was the traditional first line of treatment until the introduction of radiotherapy in the late 20th century [2–4]. The Collaborative Ocular Melanoma Study (COMS) showed that ocular radiotherapy and enucleation had similar survival rates for medium-sized tumors [5]. While radiotherapy offers an alternative for patients wanting to maintain vision and eye function, radiation-induced injury to ocular structures is a common iatrogenic consequence [4, 6–8]. Radiation causes retinal vascular endothelial cell damage and capillary closure, resulting in poor capillary perfusion [9]. Capillary nonperfusion leads to retinal ischemia, thereby promoting neovascularization through growth factors such as vascular endothelial growth factor (VEGF). These

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pathologic changes can result in neovascularization, macular edema, and vitreous hemorrhage, which can ultimately lead to tractional retinal detachment, vision loss, and neovascular glaucoma [9–12]. In the COMS, despite excluding tumors with high risk of vision loss (peripapillary tumors and large tumors), patients who received ocular brachytherapy (43%) were predicted to have a visual acuity (VA) of 20/200 or less after 3 years [13].

Radiation retinopathy (RR) is a dose-dependent ocular complication of the retina with variable latency following exposure to ionizing radiation, first described in 1933 by Stallard [3]. Although not an identical mechanism, the vascular damage and secondary ischemia in RR pathogenesis does appear to share some similar mechanisms as patients with diabetic retinopathy and retinal vein occlusions [10, 12, 14, 15]. In a study of 74 untreated eyes with uveal melanoma, the median VEGF-A concentration was 146.5 pg/mL, significantly higher than control levels of 50.1 pg/mL [16]. Furthermore, elevated VEGF has been reported in patients with untreated and pre-radiation uveal melanoma who develop secondary complications such as neovascularization compared to patients undergoing cataract removal procedures [16, 17]. Due to the inherently high levels of VEGF with uveal melanoma and the vascular damage and ischemia secondary to radiation, the focus of treatment for RR has shifted to intravitreal anti-VEGF agents. Intravitreal bevacizumab, an anti-VEGF agent, has been reported in reducing and regressing retinal edema, hemorrhages, hard exudates, and neovascularization following radiotherapy [18–21].

Bevacizumab and ranibizumab, an antibody from bevacizumab, are extensively used with reported efficacy in treatment for RR [19, 22]. Prophylactic bevacizumab injections every 4 months after plaque brachytherapy for uveal melanoma have been shown to reduce rates of optical coherence tomography (OCT)-evident macular edema and poor VA over a 2-year period compared to control patients not receiving prophylactic therapy [23]. Presently, intravitreal bevacizumab is the most commonly used treatment for RR due to cost-effectiveness and efficacy. It works by inhibiting biologically active isoforms of VEGF-A. Aflibercept is a human recombinant fusion protein that inhibits all biologically active isoforms of VEGF-A, VEGF-B, and placental growth factor. Clinical trials have validated that intravitreal aflibercept is effective in treating diabetic macular edema and macular edema secondary to central retinal vein occlusion [24, 25]. Though there is increasing support for intravitreal aflibercept in macular edema, there is a deficiency in literature on the efficacy of aflibercept in patients with RR.

Materials and Methods

The objective of this study is to determine the clinical efficacy of intravitreal aflibercept for radiation maculopathy secondary to I-125 plaque brachytherapy for uveal melanoma in those that failed intravitreal bevacizumab treatment. This study was approved by the Health Research and Ethics Board of Alberta. Initial entry criteria for this study included I-125 plaque brachytherapy treatment for uveal melanoma and the development of radiation-induced maculopathy (edema). Each patient was treated with I-125 brachytherapy with a minimum dose of 70 Gy to the apex, accounting for the silastic and reflection off the gold implant. All patients had undergone a full ophthalmic examination on each visit, including best corrected VA assessed with their current correction and pinhole measured with a ETDRS chart (BCVA), intraocular pressure, slit lamp examination, OCT of the fovea, and color photos of the fovea and tumor. Statistical analysis was performed using the logMAR score [26]. Anatomical outcomes were assessed through measuring central foveal thickness (CFT) by OCT. Both BCVA and CFT were measured before and after the switch to aflibercept. A clinically significant improvement in BCVA was defined as an increase in 1 or more lines, equivalent to 0.1 logMAR units; clinically significant reduction in CFT was defined as 100 μm. Additional pertinent patient information including age, sex, and other demographic data was collected and included in the analysis.

Eligible patients were initially placed on monthly doses of bevacizumab (1.25 mg/0.05 mL) at least 6 months prior to starting this study and then converted to monthly aflibercept (2.0 mg/0.05 mL) due to treatment failure. Failure is defined as incomplete resolution of retinal fluid despite receiving monthly intravitreal bevacizumab injections. Exclusion criteria included other intravitreal treatment prior to the switch, recent ophthalmic surgery and/or procedures, significant unrelated retinal pathology, epiretinal membrane, and incomplete patient records.

Linear regression was used to determine if the dependent variables, CFT and BCVA, were associated with independent variables including age, gender, radiation dose to fovea and optic nerve, tumor height, maximal tumor basal dimension, and distance of tumor to fovea and optic nerve. Statistical significance was defined as p value <0.05. All statistical analysis was performed on STATA (version 12, eighth edition; StataCorp LLC., College Station, TX, USA).

Results

Thirty eyes of 30 patients were enrolled from two medical centers in Canada. Seventeen females and thirteen males participated in this study and the mean age was 57 ± 15 years. No patients were excluded based on the exclusion criteria. All thirty eyes had recalcitrant radiation maculopathy and at least six monthly bevacizumab injections prior to the switch to aflibercept. The mean BCVA prior to conversion from bevacizumab to aflibercept was 20/115 (0.76 logMAR ± 0.50). The mean CFT prior to conversion was 476 ± 170 μm (Table 1).

Table 1. Participant demographics at baseline, prior to treatment conversion

Total patients, *n*	30
Total eyes, *n* (%)	30 (100)
Male, *n* (%)	13 (43)
Mean age, years	56.9±15.1
Mean baseline CFT, µm	476±170
Mean baseline LogMAR VA (Snellen)	0.76±0.5 (20/115)

LogMAR, logarithm of minimum angle of resolution.

Table 2. Characteristics following treatment switch from bevacizumab to aflibercept

CFT, mean, µm	
Baseline	476±170
1 month post aflibercept[a]	403±137, $p = 0.003$
3 months post aflibercept[a]	389±151, $p = 0.004$
6 months post aflibercept[a]	386±139, $p = 0.003$
LogMAR VA, mean (Snellen)	
Baseline	0.76±0.5 (20/115)
1 month post aflibercept[a]	0.73±0.4 (20/107), $p = 0.020$
3 months post aflibercept[a]	0.74±0.4 (20/109), $p = 0.004$
6 months aflibercept[a]	0.75±0.4 (20/112), $p = 0.013$

LogMAR, logarithm of minimum angle of resolution. [a] Compared to baseline.

At 1 month after one injection of intravitreal aflibercept, the average CFT was 402 ± 137 µm. Following 6 months of monthly intravitreal aflibercept injections, the CFT decreased at each time point, and the mean BCVA improved (Table 2).

A linear regression analysis was conducted to assess the impact of aflibercept on the change of CFT and logMAR (Fig. 1). A statistically significant reduction in CFT compared to baseline was noted at 1 month (p value = 0.003), 3 months (p value = 0.004), and 6 months (p value = 0.003) after switching to monthly aflibercept injections. Additionally, a similar pattern was seen with VA, which was significantly associated with improvement in response to aflibercept at 1 month (p value = 0.020), 3 months (p value = 0.004), and 6 months (p value = 0.013).

From baseline, 47% of patients (14/30) showed a clinically significant improvement in CFT (reduction of a minimum of 100 µm), and 23% of patients (7/30) showed a clinically significant improvement in VA (improvement of 1 line of ETDRS chart) following 6 months of intravitreal aflibercept. Ten percent of patients improved by 3 lines or more following 6 months of treatment (3/30). Further sub-group linear regression analysis was completed, comparing patients with clinically significant improvement in either CFT or VA with the remaining subjects in the study. No statistically significant baseline characteristics were associated with either clinically significant improvement: gender and CFT (p value = 0.27), age and CFT (p value = 0.21), gender and acuity (p value = 0.90), and age and acuity (p value = 0.30). Yet, CFT reduction at 1 month was statistically significantly associated with CFT at 6 months (p value <0.04). Exploratory analysis revealed that all patients who noted an improvement of CFT by 120 µm or more at 1 month retained their clinically significant reduction in CFT (7/7). But for those who noticed a smaller improvement or worsening at 1 month, only 32% had a clinically significant improvement of CFT at 6 months (7/22). In contrast, clinically significant improvement in VA at 6 months was not associated with VA change at 1 month ($p = 0.33$) or reduction in CFT at 1 month ($p = 0.69$). Reduction in CFT was not statistically associated with age (p value = 0.66), gender (p value = 0.43), dose to fovea (p value = 0.32), dose to optic nerve (p value = 0.97), tumor height (p value = 0.11), maximal basal dimension (p value = 0.07), distance of tumor edge to fovea (p value = 0.61), distance of tumor edge to optic nerve (p value = 0.77). Similarly, improvement in VA, was not statistically associated with age (p value = 0.54), gender (p value = 0.98), dose to fovea (p value = 0.43), dose to optic nerve (p value = 0.25), tumor height (p value = 0.15), maximal basal dimension (p value = 0.51), distance of tumor edge to fovea (p value = 0.92), distance of tumor edge to optic nerve (p value = 0.69). No cases of severe adverse reactions to aflibercept such as endophthalmitis, retinal tear, retinal detachment, increased intraocular pressure, or vitreous hemorrhage were observed in this study.

Discussion and Conclusion

The treatment of radiation retinopathy is an evolving area of clinical research which implicates, at least partially, increased levels of intraocular VEGF. Despite many groups demonstrating resolving edema and symptomatic improvement of RR with intravitreal anti-VEGF, namely bevacizumab [27–31], there is no current consensus on a treatment protocol. Recently, limited information on the

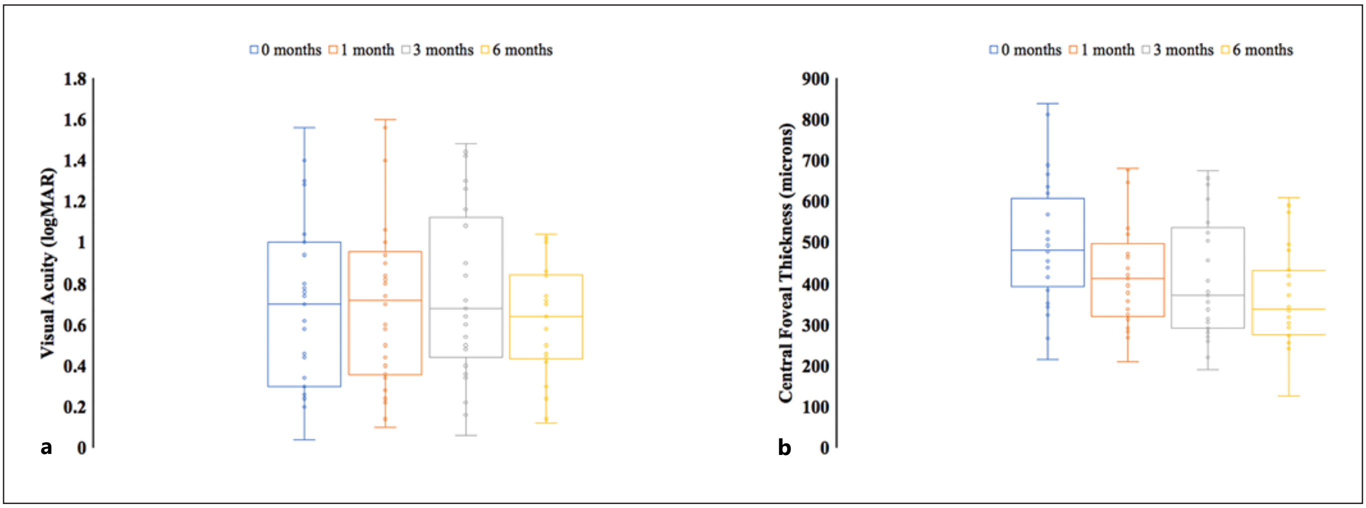

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Fig. 1. The mean change in CFT and BCVA measured by ETDRS and represented in logMAR (n = 30) during 6-month exposure to aflibercept after 6 months of prior bevacizumab treatment in patients with radiation retinopathy. Following the switch to aflibercept, the average CFT steadily dropped over the course of 6 months (**a**), whereas VA slight improved and then slightly rebounded (**b**). Changes in from 1 month, 3 months, and 6 months to baseline were significantly correlated in both CFT and VA. The top and bottom whiskers indicate the maximum and minimum values, respectively.

use of intravitreal aflibercept has also shown anatomical and functional benefits. Aflibercept is a human recombinant fusion protein that is extensively used for diabetic macular edema, macular degeneration, central and branch retinal vein occlusions, and other sources of neovascularization within the eye [32–35].

Four studies have documented improvement with aflibercept as a primary treatment for radiation retinopathy. In 2017, Pooprasert et al. [35] demonstrated improvement in radiation retinopathy in a single case of radiation retinopathy. In 2019, Karagiannis et al. [32] utilized primary intravitreal aflibercept and demonstrated improvement in a patient with bilateral peripheral radiation retinopathy and unilateral maculopathy presumed to be related to external beam radiotherapy of an optic nerve glioma. Fallico et al. [33], in 2019, utilized intravitreal aflibercept as a primary treatment of 9 patients with radiation maculopathy secondary to treatment of uveal melanoma with ruthenium-106. They found a statistically significant improvement in vision and CFT without any complications. In 2019, Murray et al. [34] published a prospective clinical study of 40 patients with radiation maculopathy randomized to either five/six weeks of aflibercept or treat and adjust as primary treatment. They concluded that the maculopathy improved equally with both treatment algorithms.

Two studies on age-related macular degeneration have shown benefit from switching to aflibercept as a second-line treatment in patients that have an insufficient response to other intravitreal anti-VEGF agents such as bevacizumab [36, 37]. There have only been a handful of case-studies and one retrospective case series totaling 6 patients to date in RR showing improvement with aflibercept in patients with post-radiation macular edema who had seen little to no benefit with bevacizumab [38, 39]. Khan et al. [39] presented a case-controlled series with 5 patients switched to aflibercept and five remaining on bevacizumab whom were previously treated with bevacizumab for their resistant cystoid macular edema. In comparison to the control group, the aflibercept group had significantly lower macular thickness and improved VA. Central macular thickness decreased from 463 ± 138 to 267 ± 80 μm and the mean VA improved from 20/67 to 20/42 (p = 0.03). Loukianou and Loukianou [38] presented a single patient with radiation maculopathy secondary to external beam radiotherapy who had received five and seven bevacizumab injections in the right and left eye, respectively, over the span of 1 year with no improvement. Subsequently, monthly intravitreal aflibercept injections were administered in both eyes, and both central retinal thickness and (VA) demonstrated marked improvements. However, it appears that the bevacizumab injections the patient was previously administered were not monthly, as the aflibercept injections were, which could have played a role in the results of the case report. These early results in 6 cases suggest a possible role for aflibercept in patients not responding to bevacizumab.

In this prospective multicentre study, intravitreal aflibercept was used as a rescue therapy for 30 patients with radiation retinopathy that had failed 6 months or more of monthly intravitreal bevacizumab treatment. The mean CFT improved from 476 ± 170 to 386 ± 139 µm over 6 months (p value = 0.003, Table 2). The mean BCVA improved minimally from 20/115 ± 20/63 to 20/112 ± 20/54 (p value = 0.013) over the course of 6 months (Table 2). Overall, in our study, after 6 months of intravitreal aflibercept, 46% of patients displayed clinically significant anatomical improvements (a CFT improvement of 100 µm or greater) and 23% of patients showed improvement in BCVA (1 line or better). Exploratory analysis demonstrated that if a patient demonstrated an improvement of 120 µm or more at 1 month after receiving their first dose of aflibercept, they were 100% likely to have a clinically significant improvement of CFT at 6 months, yet only 32% who did not demonstrate this level of improvement at 1 month demonstrated a 100 µm improvement at 6 months. We evaluated age, gender, radiation dose to fovea and optic nerve, tumor height, maximal tumor basal dimension, and distance of tumor to fovea and optic nerve in all patients, but these were all statistically insignificant in trying to predict reduction in CFT or improvement in VA. Therefore, based on this small case series, the only way to improve VA and CFT in patients who have failed bevacizumab is to trial the use of aflibercept, and early response at 1 month appears to be the best predictor of long-term response.

The main limitations of this study are the small sample size and the lack of a control group. However, it is the largest study evaluating aflibercept in the context of RR-related macular edema for patients who had previously failed bevacizumab therapy to date. Furthermore, chronic cystoid macular edema, prior to switching to aflibercept, may have limited our ability to achieve improvements in VA. RR is a rare disease [40–42], and studying it is an arduous process given the multifactorial development of vision loss and radiation retinopathy, and that any loss of VA or anatomical function cannot solely be attributed to reversible radiation damage. Nevertheless, the results we have provided are representative of real-world encounters that are in keeping with other ophthalmologists that would be treating this subset of patients. Future studies will benefit from expanding beyond two sites, initiating rescue treatment earlier in the disease course, collecting data beyond the 6-month time frame, and comparing aflibercept to ranibizumab, sub-Tenon's triamcinolone, or intravitreal injection of triamcinolone or dexamethasone in patients with RR secondary to uveal melanoma.

In conclusion, preliminary research on intravitreal aflibercept injections suggests it is a safe treatment that could be considered in patients with radiation retinopathy following an incomplete response to first-line management by intravitreal bevacizumab. Our study demonstrated that 47% of patients had a clinically significant improvement in CFT, and 23% had a clinically significant improvement in vision. Randomized clinical trials across multiple sites and control groups are needed to further confirm the efficacy of the switch from bevacizumab to aflibercept in patients with radiation retinopathy.

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Statement of Ethics

This study protocol was reviewed and approved by the Health and Ethics Research Board of Alberta (HERBA), which is the institute's committee on human research. The approval number was HREBA.CC-17-0625. Written informed consent was obtained from all participants in the study.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

Ojas Srivastava – original draft writing, analysis, and reviewing. Ezekiel Weis – conceptualization, methodology, revision, and supervision.

Data Availability Statement

All data generated or analyzed during this study are included in this article. Further inquiries can be directed to the corresponding author.

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